Ceramic composite materials with MoS2 and SiO2

"Which ceramic matrices can host MoS2 nanosheets with SiO2?" *Generated by LeapSpace*

The ceramic matrices directly supported by the retrieved evidence are *porous Si_3N_4*, *TiO_2*-based ceramic structures, and *$H_2Ti_2O_5$* titanate; *SiO_2* itself is also used as the matrix or shell around MoS_2, but in those cases it is the ceramic host rather than an extra third phase. [90] [91] [92] [93] [94] [95] [96]

Ceramic hosts that are supported

Ceramic matrix / host	How it hosts MoS_2 with SiO_2	What the evidence shows
Porous Si_3N_4 ceramic	MoS_2 is grown **inside the pores** of porous Si_3N_4, while SiO_2 is part of the surrounding silica-based composite architecture [90]	This is the clearest direct example of a ceramic matrix hosting MoS_2 with silica present [90]
TiO_2-based ceramic structures	MoS_2 is combined with TiO_2 in several MoS_2/TiO_2 systems, and one ternary composite explicitly uses **$MoS_2/SiO_2/TiO_2$** [91] [95] [95]	TiO_2 is a supported ceramic platform for MoS_2, and it can be part of a SiO_2-containing ternary ceramic composite [91] [95]
$H_2Ti_2O_5$ titanate	A ternary **$H_2Ti_2O_5$@MoS_2@SiO_2** core-shell material was prepared, with the titanate serving as the inner ceramic core [92] [96]	This directly supports $H_2Ti_2O_5$ as a ceramic host for MoS_2 and SiO_2 in a hierarchical composite [92] [96]
SiO_2 matrix / shell	MoS_2 is embedded in or coated by SiO_2 in multiple core-shell and nanocomposite systems, including **MoS_2@SiO_2** and **Pt-MoS_2/SiO_2** platforms [93] [94] [96]	SiO_2 is clearly used as the ceramic host layer or matrix around MoS_2 nanosheets [93] [94] [96]

Bottom line

If you want **ceramic matrices that can host MoS_2 nanosheets together with SiO_2**, the strongest directly supported choices are **porous Si_3N_4**, **TiO_2-based ceramics**, and **$H_2Ti_2O_5$ titanate**; **SiO_2** is also repeatedly used as the host/shell itself in MoS_2–SiO_2 composites. [90] [92] [93] [94] [95]

Confidence level: **Medium** — the evidence directly supports several concrete ceramic hosts, but it does not provide a comprehensive survey of all possible ceramic matrices.

References

90. Bai, J., Huang, S., Yao, X., (...), Huang, Z. (2023) Surface engineering of nanoflower-like MoS2 decorated porous Si3N4 ceramics for electromagnetic wave absorption. *Journal of Materials Chemistry A*. https://doi.org/10.1039/d3ta00122a

91. Santalucia, R., Vacca, T., Cesano, F., (...), Scarano, D. (2021) Few-layered MoS2 nanoparticles covering anatase TiO2 nanosheets: Comparison between ex situ and in situ synthesis approaches. *Applied Sciences (Switzerland)*. https://doi.org/10.3390/app11010143

92. Chen, Y., Sun, W., Zheng, H., (...), Hao, C. (2021) Electrorheological response behavior of H2Ti2O5@MoS2@SiO2 core-shell nanoparticles. *Ceramics International*. https://doi.org/10.1016/j.ceramint.2021.05.118

93. Kwok, K.M., Ong, S.W.D., Chen, L., Zeng, H.C. (2018) Constrained Growth of MoS2 Nanosheets within a Mesoporous Silica Shell and Its Effects on Defect Sites and Catalyst Stability for H2S Decomposition. *ACS Catalysis*. https://doi.org/10.1021/acscatal.7b03123

94. Zhu, X., Xue, Y., Hou, S., (...), Liu, Q. (2023) Highly selective colorimetric platinum nanoparticle-modified core-shell molybdenum disulfide/silica platform for selectively detecting hydroquinone. *Advanced Composites and Hybrid Materials*. https://doi.org/10.1007/s42114-023-00719-z

95. Steffi, A.P., Ramachandran, R., Chen, S.-M., (...), Narendhar, C. (2021) Rational Construction of SiO2/MoS2/TiO2 Composite Nanostructures for Anti-Biofouling and Anti-Corrosion Applications. *ChemistrySelect*. https://doi.org/10.1002/slct.202004263

96. Luo, D., Peng, T., Liu, S., (...), Chen, C. (2025) Ultrasensitive and selective nanomolar detection of quercetin in 12 complex food matrices via 3D hierarchical SiO_2@MoS_2/MXenes heterostructured sensor. *Food Chemistry*. https://doi.org/10.1016/j.foodchem.2025.144462